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2D MAMMOGRAPHY INTERPRETATION, DIAGNOSTIC PERFORMANCE COMPARING ARTIFICIAL INTELLIGENCE WITH RADIOLOGISTS



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BACKGROUND

- Breast cancer remains the most frequently diagnosed cancer among Malaysian women (1).
- 2D mammography depends heavily on radiologist experience, reader variation and daily workload pressures (2).
- Artificial intelligence (AI) has shown potential to support mammographic interpretation (3,4,5).
- This study evaluated diagnostic performance of AI and compared it with radiologists.

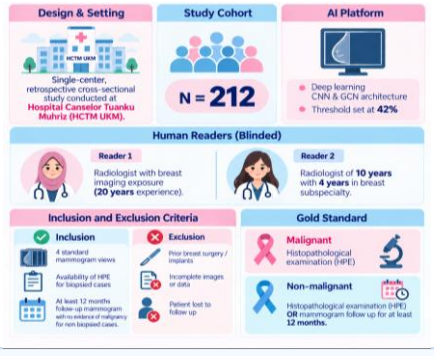
OBJECTIVES

Primary objective: To evaluate stand-alone diagnostic performance of AI for breast cancer detection on 2D mammography.

Secondary objective: To compare diagnostic performance and agreement between AI and two radiologists

Outcomes: Sensitivity, specificity, PPV, NPV, AUC and Cohen's Kappa (κ) agreement

METHODS



RESULTS

Comparison	Agreement (%)	Cohen's Kappa (κ)	95% CI	Interpretation
Reader 1 vs Reader 2	86.8	0.729	0.637–0.822	Substantial
Reader 1 vs AI	79.2	0.585	0.478–0.692	Moderate
Reader 2 vs AI	77.4	0.547	0.435–0.659	Moderate

Table 1. Pairwise agreement between the two readers and AI for breast cancer classification

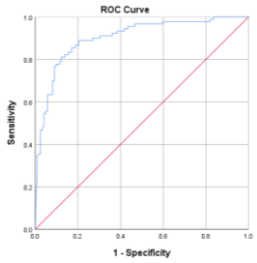


Figure 1: Receiver Operating Characteristic (ROC) curve of AI system

Diagnostic performance	AI	Reader 1	Reader 2
Sensitivity (%)	90.0	84.4	91.1
Specificity (%)	79.5	93.4	90.2
PPV (%)	76.4	90.5	87.2
NPV (%)	91.5	89.1	93.2

Table 2. Comparison of diagnostic performance between AI and two readers

- AI has high sensitivity but lower specificity than radiologists.
- Reader 1 has highest specificity.
- Reader 2 showed the highest sensitivity and NPV.
- R1 and R2 showed substantial agreement, while agreement between AI and each reader is moderate.
- ROC curve demonstrated an excellent discriminative ability of AI (AUC=0.903, p-value < 0.001)

CONCLUSION / TAKE-HOME MESSAGE

1. Main finding

AI demonstrated high sensitivity (90%) and an excellent AUC (0.903) performing comparably to radiologists in detecting breast cancer.

2. Clinical meaning

AI's high sensitivity minimizes missed cancers (false negative), while radiologist interpretation remain vital to offset AI's lower specificity and reduce false-positive recall rates.

3. Next step

AI can function as reliable triage tool or second reader in high-volume setting, reducing reader fatigue and radiologists' workload.

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